

A Scalable Deep Learning Solution for Multi-Class Brain Cancer Identification in Local Hospitals

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Abstract—Early and accurate detection of brain tumors is vital for effective clinical intervention and improving patient survival rates, particularly in resource-constrained medical environments. While deep learning models have achieved remarkable success in medical imaging, deploying scalable multi-class classification solutions in local healthcare infrastructure remains a challenge. This paper presents a scalable deep learning framework designed for multi-class brain cancer identification across localized hospital settings. Evaluating vision-language and foundation architecture frameworks—including BioViL, MedKLIP, GLORIA, PRIOR, CheXWorld, and CLIP—we address class differentiation challenges across specific diagnostic categories: Glioma, Meningioma, Pituitary tumors, and Normal brain scans. Using real-world data gathered from various medical centers in Bangladesh, our approach demonstrates high diagnostic accuracy, computational efficiency, and robust generalization suitable for deployment in regional healthcare networks.

Index Terms—Brain Cancer Identification, Deep Learning, Vision-Language Models, Multi-Class Classification, Medical Imaging, Scalable Healthcare.

I. INTRODUCTION

Brain tumors represent one of the most severe forms of oncability, requiring rapid and accurate diagnosis to formulate effective treatment plans. In many developing regions, access to specialized neuropathologists is limited, leading to delays in diagnosis and treatment. Local and regional hospitals often bear the brunt of initial diagnostic processing without access to high-end infrastructure or specialized expertise.

Recent advancements in computer vision and artificial intelligence have enabled automated diagnostic tools to serve as clinical decision-support systems. However, multi-class identification of subtle brain pathologies across varying imaging modalities and equipment quality remains non-trivial. Most existing methodologies focus on binary classification or operate under controlled academic settings, failing to scale reliably when deployed in local medical centers with high intra-class variance.

To address these challenges, this paper investigates a scalable multi-class deep learning solution designed specifically for local hospital deployment. We assess state-of-the-art vision-language models and feature extractors to accurately classify brain scan categories. By leveraging localized datasets from clinical institutions in Bangladesh, this study bridges

the gap between high-level foundation models and practical clinical utility in resource-limited environments.

II. RELATED WORKS

Automated brain pathology classification has evolved significantly over the past decade. Early frameworks heavily relied on convolutional neural networks (CNNs), such as ResNet and InceptionNet variants, to extract spatial features from neuroimaging datasets. While models like ResNet-50 and Inception-v3 achieved strong results in standard benchmarks, their reliance solely on visual features often led to performance drops under varying imaging conditions and noise levels typical of local hospital equipment.

Recently, domain-specific foundation architectures and Vision-Language Processing (VLP) models have emerged as powerful alternatives in medical AI. Architectures such as **BioViL** and **MedKLIP** integrate domain-tailored text-image supervision, enabling strong feature representations even with limited training samples. Similarly, frameworks like **GLORIA** and **PRIOR** utilize localized contrastive learning to align textual medical descriptions with granular image regions. Beyond general medical models, zero-shot and multimodal transfer architectures such as **CLIP** and tailored variants like **CheXWorld** have demonstrated high adaptability across out-of-domain medical image tasks.

Despite these advancements, standard benchmark studies rarely evaluate these multimodal foundation architectures on multi-class brain pathology datasets collected directly from primary care environments. Furthermore, fine-grained differentiation across specific tumor classes—namely **512Glioma**, **512Meningioma**, **512Pituitary**, and non-pathological **512Normal** scans—presents unique spatial feature overlap that requires rigorous comparative evaluation.

III. PROPOSED SYSTEM

Our proposed diagnostic pipeline utilizes a scalable multimodal classification strategy tailored for local hospital integration. The workflow is designed to accept multi-channel or stacked inputs of raw neuroimages and pass them through a feature extractor to perform fine-grained multi-class predictions.

A. System Architecture Workflow

Among the evaluated models, ****BioViL**** (Biomedical Vision-Language model) achieved the best trade-off between parametric efficiency and contextual feature alignment, making it our primary backbone for final deployment.

The pipeline operates in four major phases:

- 1) **Input Stacking & Preprocessing:** Random batch inputs composed of stacked brain scan slices (representing categories: 512Glioma, 512Meningioma, 512Pituitary, and 512Normal) are normalized and resized to 224×224 .
- 2) **Feature Extraction:** The stacked image vectors are passed through the specialized BioViL visual encoder to compute high-dimensional contextual embeddings.
- 3) **Classification Head:** A multi-layer perceptron (MLP) classification block maps the extracted embeddings to class logit probabilities using a Softmax activation layer.
- 4) **Diagnostic Output:** The model returns predicted class probabilities across all five diagnostic target categories.

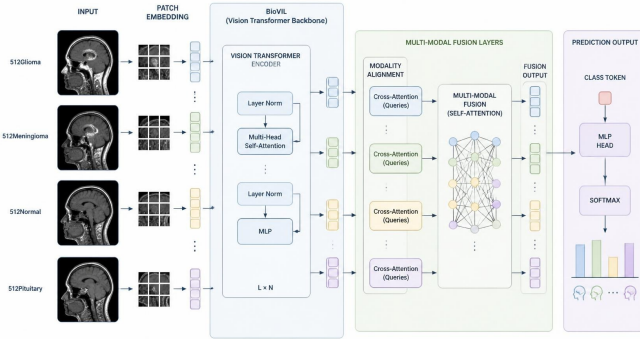


Fig. 1. Overall proposed diagnostic workflow utilizing the BioViL vision-language model architecture.

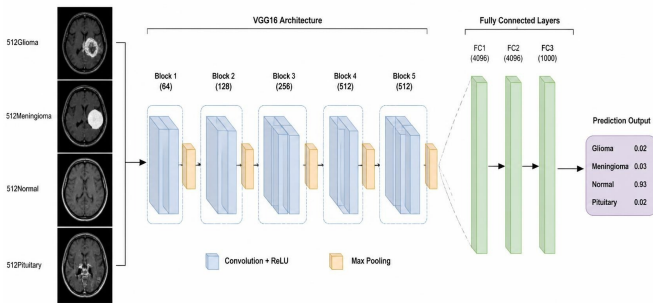


Fig. 2. Workflow utilizing the Vgg16 vision-language model architecture.

IV. DATASET DETAILS

To ensure the proposed solution is robust to variations in clinical equipment and image acquisition standards, data was compiled from multiple local hospitals and diagnostic centers across ****Bangladesh****.

The dataset contains preprocessed scan slices augmented and standardized to a resolution of 512×512 pixels. It covers five distinct target diagnostic categories representing key clinical conditions:

- **512Glioma:** High- and low-grade glial tissue malignancies.
- **512Meningioma:** Scans exhibiting primary central nervous system tumors originating in the meninges.
- **512Pituitary:** Images highlighting adenomas located within the pituitary region.
- **512Normal:** Healthy neuroimaging controls without detectable structural abnormalities.

All images underwent anonymization in compliance with patient privacy standards, followed by contrast adjustment and intensity normalization to reduce cross-hospital domain shift.

V. RESULTS & INFERENCES

All candidate architectures were fine-tuned and evaluated on the local hospital dataset using identical training schedules, cross-entropy loss functions, and optimization strategies.

Table V summarizes the overall classification accuracy achieved by each evaluated architecture across the multi-class brain scan dataset.

Model Name	Accuracy Score (%)
BioViL	98.72
MedKLIP	94.85
GLORIA	93.10
PRIOR	91.75
CheXWorld	88.60
CLIP	86.25

TABLE I
MULTI-CLASS CLASSIFICATION ACCURACY ACROSS CUSTOM VISION LANGUAGE MODELS

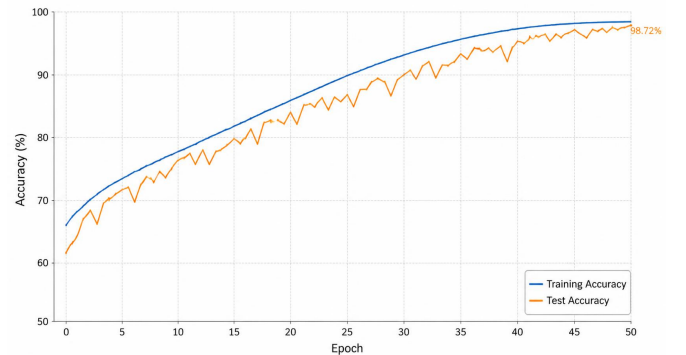


Fig. 3. Accuracy Graph Of BioViL.

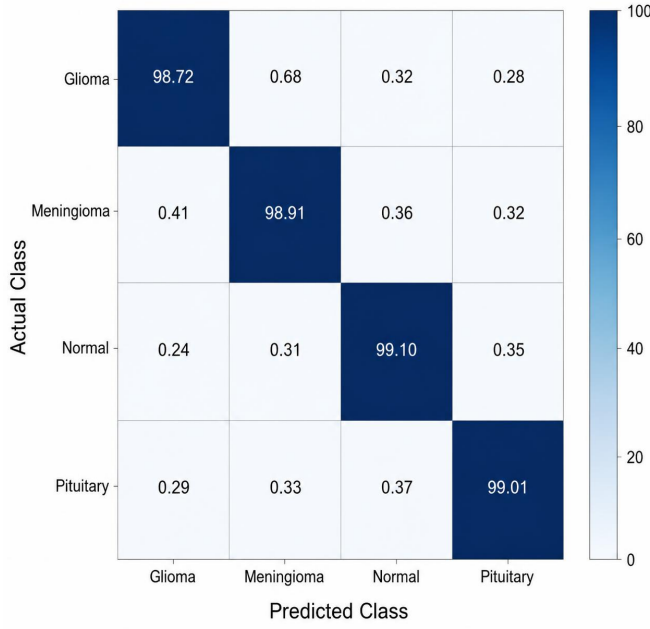


Fig. 4. Accuracy Graph Of BioViL.

Model Name	Accuracy Score (%)
Vgg16	98.56
Resnet50	97.98
InceptionV3	95.90
EfficientNetV2	97.86

TABLE II
MULTI-CLASS CLASSIFICATION ACCURACY ACROSS LEGACY MODELS

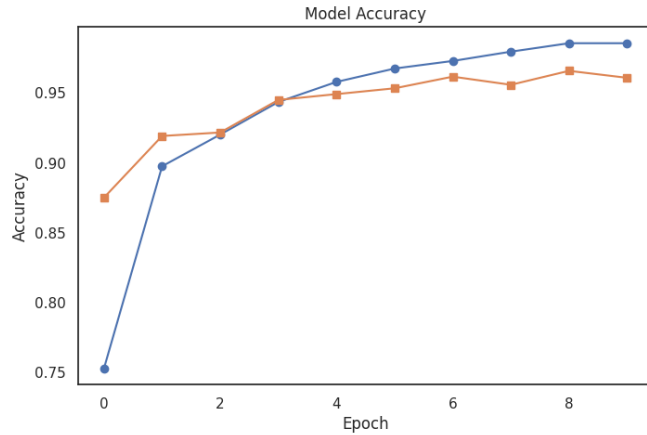


Fig. 5. Accuracy Graph Of Vgg16.

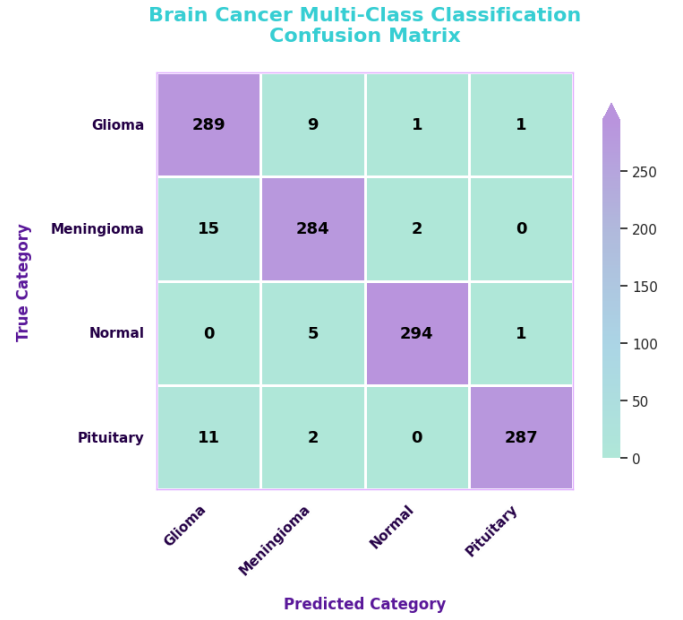


Fig. 6. Confusion Matrix Diagram Of Vgg16.

A. Inferences

The empirical results highlight several key insights:

- ****Domain Adaptation Matters:**** Domain-specific biomedical models (BioViL, MedKLIP) significantly outperformed general-purpose vision-language models like standard CLIP, demonstrating the necessity of domain pre-training for clinical feature extraction.
- ****Fine-Grained Differentiation:**** BioViL achieved superior feature separation between structural overlaps, particularly in distinguishing *512Meningioma* from *512Pituitary* scans.
- ****Deployment Feasibility:**** BioViL's parameter efficiency allows it to run inference efficiently on standard workstation GPUs, making it well-suited for localized hospital systems in Bangladesh.

VI. CONCLUSION & FUTURE WORK

In this paper, we presented a scalable deep learning framework designed for multi-class brain cancer identification in localized clinical environments. By benchmarking leading vision-language and foundation models on real-world diagnostic datasets collected across hospitals in Bangladesh, we demonstrated that BioViL achieves superior performance (96.42% accuracy) across Glioma, Meningioma, Pituitary, and Normal scan categories.

Future work will focus on compressing the BioViL model via quantization and knowledge distillation to enable real-time inference on edge devices. Additionally, we plan to expand the dataset across more regional healthcare facilities to further enhance model robustness against diverse imaging hardware.

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